

A Propagation Prediction Method for Non-recurrent Traffic Congestion

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Abstract—Predicting non-recurrent congestion propagation patterns in the near future is crucial for GPS navigation systems to provide drivers with route recommendations for bypassing congested roads in real time. However, it is difficult to model non-recurrent congestion propagation due to the irregularity of propagation patterns and lack of sufficient relevant non-recurrent traffic data corresponding to a given traffic network. These limitations make it difficult to develop practical methods that can be applied to real traffic incidents in an online manner. To overcome this difficulty, we propose a novel online learning method using pattern matching techniques to predict the propagation patterns of non-recurrent traffic congestion. The proposed method assumes that current non-recurrent propagation patterns are similar to historical recurrent patterns. We validated the proposed method and underlying assumptions using 215 real traffic incidents and related user trajectory data obtained from a major GPS navigation system in South Korea. A comparison study is carried out with the existing benchmarks, which revealed a superior prediction performance and robustness of the proposed method.

Index Terms—Non-recurrent congestion propagation, pattern matching, dynamic time warping (DTW), online learning

I. INTRODUCTION

Traffic congestion has become a universal problem in urban cities based on dramatic population growth. Corresponding increases in vehicles, infrastructure, and the proliferation of delivery services have also contributed to this problem. Congestion can cause various social, environmental, and economic problems by reducing the efficiency of transportation infrastructure and increasing travel time, air pollution, and fuel consumption.

Traffic congestion caused by specific events, rather than by regular traffic congestion during rush hours, is called non-recurrent traffic congestion [1]. This type of congestion occurs as a result of unexpected events such as vehicle collisions, road maintenance construction, or debris. Among these types of non-recurrent congestion, traffic incidents are the most difficult to predict, but must be handled in a timely and effective manner to alleviate the impact on road capacity and time loss, and to reduce the likelihood of secondary incidents [2].

Predicting non-recurrent propagation patterns in an online manner is especially important for navigation service providers because they must provide new routes for drivers who are in or heading toward an incident area to help them avoid congested roads. However, it is challenging to model non-

recurrent congestion propagation using typical machine learning methods due to the irregularity of speed patterns during incidents and lack of training incident data for a given area. These characteristics do not align with the typical assumptions adopted in time series modeling, such as the stationarity or abundance of a dataset [3]. Therefore, although numerous attempts have been made to predict recurrent propagation patterns (e.g., rush hour traffic) [4], [5], there have been few attempts to predict propagation patterns after non-recurrent events such as traffic incidents.

In contrast, “pattern matching” does not require any assumptions and is based on the imitation of historical patterns. Pattern matching methods identify historical patterns and assume that the patterns in current circumstances will evolve in a similar manner to the identified patterns. In this study, by utilizing the concept of pattern matching, we developed a novel method that predicts non-recurrent congestion propagation patterns by identifying recurrent congestion propagation patterns that have occurred in the past. The underlying assumption of the proposed method is that current non-recurrent propagation patterns will be similar to historical recurrent congestion patterns if the speed patterns of an incident road in the present and past are similar in scenarios where traffic network topology conditions are constant. We use dynamic time warping (DTW) to match the patterns of speed time series between the past and present by measuring the similarity between two time series.

Additionally, the proposed method enables incident-based online learning. Online learning refers to the process of updating a model as new data such as different traffic network regions or new speed patterns arrive over time. Traditional machine learning and deep learning methods fix specific traffic networks and train models only for those regions, which limits the learning of new locations and time periods [6]. Considering that incidents occur spontaneously in multiple areas and must be monitored over time, such methods are not suitable for real-time scenarios. However, our model can be applied in real-world scenarios anywhere and anytime because it performs point-in-time pattern matching for individual incidents.

The main contributions of our work can be summarized as follows:

- We propose a novel online learning method based on pattern matching to predict the propagation patterns of non-recurrent traffic congestion.

- We hypothesize and demonstrate that historical recurrent congestion pattern information is useful for predicting real-time non-recurrent congestion patterns.
- We validate the proposed method using 215 real traffic incidents and related user trajectory data obtained from a major GPS navigation system in South Korea.

The remainder of this paper is organized as follows. Section II reviews related works on general approaches to recurrent congestion propagation prediction and pattern matching used for congestion analysis. Section III proposes a novel methodology for predicting the propagation patterns of non-recurrent traffic congestion. Section IV presents related experimental results that justify the application of the proposed model to a real-world navigation system. Finally, the conclusions of this study are summarized in Section V.

II. RELATED WORKS

A. Congestion propagation

Related studies on propagation patterns have mainly focused on identifying frequently recurring congestion propagation footprints in the past. [7] proposed the STOTree algorithm and a frequent subtree algorithm for finding a frequent propagation tree of abnormal traffic flow between regions. [8] extended this work to develop a visual interface to explore traffic congestion and propagation patterns. [4] predicted the footprints of recurrent congestion propagation patterns in the form of propagation graphs called Pro-Graphs. [9] compared various traffic congestion prediction methods using processed GPS trajectory data and concluded that deep learning models obtain higher accuracy compared with conventional machine learning models. However, none of these studies have attempted to predict non-recurrent congestion propagation.

The other type of research related to our work is the prediction of near-future traffic conditions based on non-recurring events. [10] utilized the cell transmission model and proposed a simulation-based non-recurrent congestion propagation model, which was incorporated with information gained from advanced traveler information systems (ATIS). This demonstrated that information provided by ATIS is helpful for drivers, particularly during non-recurrent congestion caused by severe incidents. [11] proposed two models to predict next-time-step traffic volumes on a single road segment to characterize congestion based on current inflow, outflow, and traffic volumes on the target road segment and adjacent upstream road segments. [12] predicted propagation speeds and boundaries based on a shockwave model for congestion following an observed traffic incident.

B. Traffic speed pattern recognition

The pattern matching and pattern imitation methods presented in [3], [13] are based on local modeling. Such methods avoid common assumptions that are required in the context of time series modeling, such as stationarity or Gaussian conditions, and are based on the imitation of patterns in historical data. They attempt to identify predefined patterns in past data histories neighboring the current time and assumes

that the current scenario will evolve in a manner similar to the historical scenario. This method has several advantages in that it does not require prior knowledge or parameter identification, making it suitable for nonlinear and uncertain systems. Additionally, it has been empirically observed that results improve with sufficient historical data [14].

Pattern matching methods with DTW have been widely used in the field of traffic for clustering congestion by comparing pattern differences between normal traffic and congested traffic. [15] estimated the traffic on neighboring roads around a congested road by comparing barometer sensor readings with a pre-stored road signature utilizing DTW. [16] proposed a trajectory pattern matching method based on DTW and spectral clustering to detect abnormal traffic flows. [17] utilized DTW distance to analyze the similarity between vehicle trajectories and clustered vehicle trajectories based on the similarity scores. However, DTW has never been applied to match patterns between non-recurrent congestion from the present and recurrent congestion from the past. In this paper, we propose a method for predicting future non-recurrent congestion propagation patterns by discovering similar patterns in historical time periods.

C. Real-time congestion prediction

The essential purpose of traffic management systems is to provide drivers with timely and reliable information regarding incidents [18]. Therefore, traffic management systems should utilize models that operate in real-time scenarios.

Traffic data are collected from streaming data sources and used for traffic modeling [19], [20]. Endless vehicular data are produced by inductive loops, GPS units in vehicles, traffic cameras, and infrared sensors. Therefore, real-time machine learning approaches should be considered in the traffic domain. Real-time machine learning (also known as stream learning or stream data mining) has acquired special relevance with the arrival of the big data era [21], [22]. Numerous attempts have been made to utilize real-time stream data. [23], [24] used real-time GPS data equipped in a floating car to detect the traffic conditions of road network. [25] implemented a GPS based road traffic congestion reporting system using Smartphone, equipped with a GPS receiver to probe congestion on the roads. [26] presented an expert system for detecting traffic congestion and incidents from real-time GPS data collected from GPS trackers or drivers' smartphones.

Compared to existing methods, we focus on the problem of congestion propagation prediction. To the best of our knowledge, this is the first attempt to develop a method of predicting congestion propagation based on GPS data in real-time scenarios.

III. METHODOLOGY

We propose a novel propagation prediction method for non-recurrent traffic congestion. Fig. 1 presents an overview of the proposed method.

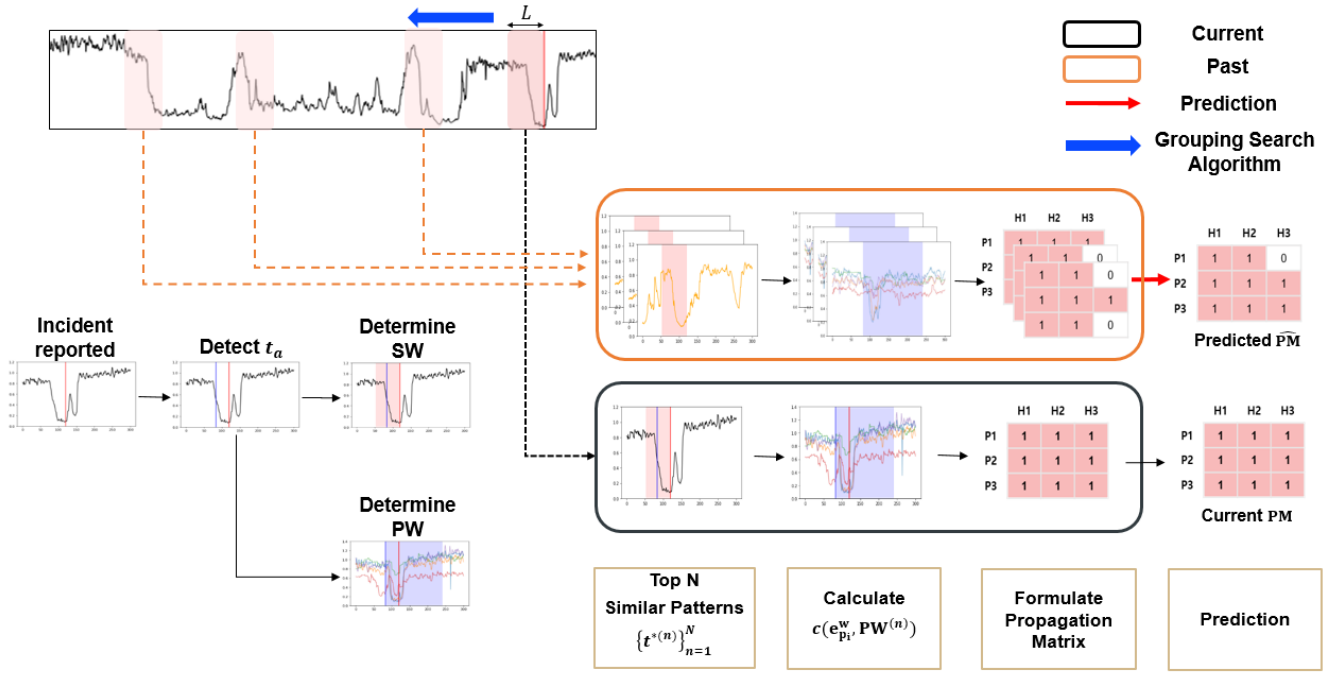


Fig. 1. Methodology overview

A. Definition of terminology

A road network $\mathcal{G} = (V, E)$ is a directed connected graph, where V is a set of nodes representing intersections and E is a set of road segments connecting nodes. Let $v_i \in V$ and $v_j \in V$ denote the nodes in $\mathcal{G}$. The ordered pair $\{v_i, v_j\}$ draws an arc in the form of an arrow from v_i to v_j , denoted as $e_{ij} = (v_i, v_j) \in E$. $v_i = \text{src}(e_{ij})$ represents the source node of the edge e_{ij} , while $v_j = \text{dst}(e_{ij})$ represents the destination node of the edge e_{ij} .

In a graph, we define an *in* operator such that the operator $\text{in} : V \rightarrow E$ returns all edges for which node v_i is the destination, such as $\text{in}(v_i) = \{e_{ki}\}_{e_{ki} \in E}$. Additionally, $|\cdot|$ is defined as the number of set elements. For example, $|\text{in}(v)|$ is the number of road segments coming into node v .

Given a graph $\mathcal{G}$, we assume that the speed information of all road segments $e \in E$ is available for a predetermined time point $t \in T$, which is denoted as $\{s(e, t) : E \times T \rightarrow \mathbb{R}^+\}$. Additionally, we let $s(e, T)$ denote the time series of speeds at road segment $e \in E$ during a predetermined time period T . Because $s(e, T)$ is a vector, we can easily define $\min(s(e, T))$, $\max(s(e, T))$, $\text{sum}(s(e, T))$, $\text{avg}(s(e, T))$, $\text{sd}(s(e, T))$, etc. (sd stands for standard deviation.) For additional analysis, we define some basic terminology as follows.

Definition 1. (Exponential moving average): Let $s(e, t)$ be the speed on traffic incident road segment e at time t . Then, the exponential moving average $\bar{s}(e, t)_n$ can be defined recursively as

$$\bar{s}(e, t)_n = \alpha \bar{s}(e, t-1)_n + (1 - \alpha)s(e, t).$$

In Definition 1, n indicates the average time period [27] and the weight factor α plays a role in controlling the importance of recent information. Although there are many possible choices for the weight factor, the most common choice is $\alpha = \frac{2}{n+1}$. The exponential moving average technique is typically applied to time series data to smooth short-term variations and capture long-term trends or cycles in the form of a moving average.

Definition 2. (k -hop incoming neighbors): We define the k -nearest hops of the set of incoming edges feeding traffic into a target edge e_0 , which are referred to as k -hop incoming neighbors, as

$$N_{in}^k(e_0) := \bigcup_{x \in N_{in}^{k-1}(e_0)} \text{in}(\text{src}(x)).$$

Definition 3. (Path, path set): Given a road network $\mathcal{G} = (V, E)$ with a set of nodes $V = \{v_1, \dots, v_n\}$ and set of edges $E \subseteq V \times V$, a path in $\mathcal{G}$ is defined as a finite sequence

$$P = (p_0 e_{p_0 p_1} e_{p_1 p_2} \dots e_{p_K p_{K+1}})$$

with $p_k \in V$ and $e_{p_k} = (p_k, p_{k+1}) \in E$ for all $k \in \{0, \dots, K\}$. For road networks, a path is uniquely determined by its edge sequence and the notation can be simplified as

$$P = (e_{p_0} e_{p_1} e_{p_2} \dots e_{p_K}).$$

A path set in $\mathcal{G}$ is defined as a set of paths whose origin and length are identical in a road network. To specify the origin and length of a path, a path set is denoted as $P^K(e_0)$, where e_0 and K are the origin and length of the path, respectively.

The order of the edge sequences is opposite to that of the traffic flows.

Definition 4. (Congested, congestion indicator function): Given an edge e , time duration T , and short-term time period $[t-d:t] \in T$, an edge e is said to be congested in T with d if $\exists t \in T$ such that

$$\max s(e, [t-d:t]) < 0.7 \times ff(e),$$

where $ff(e)$ is the free-flow speed of road e . For simplicity, we use the maximum speed restriction as the free-flow speed for road e . Then, the congestion indicator function is defined as

$$c_d(e, T) = \begin{cases} 1, & \text{if an edge } e \text{ is congested in } T \text{ with } d \\ 0, & \text{otherwise.} \end{cases}$$

For simplicity, $c_d(e, T)$ can be abbreviated as $c(e, T)$ and we assume $d = 10$.

Definition 5. (Sequence of propagation indicators, propagation indicator function): Given a path $P = (e_{p_0} e_{p_1} \dots e_{p_K})$ in $\mathcal{G}$ and a time duration T , a sequence of propagation indicators is defined as a finite sequence

$$p\text{-seq}(P, T) = (q(e_{p_1}, T), q(e_{p_2}, T), \dots, q(e_{p_K}, T)),$$

where $q(e_{p_i}, T)$ is the propagation indicator function. Assuming that $c(e_{p_0}, T) = 1$, the propagation indicator function is defined as

$$q(e_{p_i}, T) = \begin{cases} 1, & \text{if } c(e_{p_i}, T) = 1 \text{ and } c(e_{p_{i-1}}, T) = 1 \\ 0, & \text{otherwise,} \end{cases}$$

for $i \geq 1$.

B. Problem statement

We consider a road e_0 that is reported to have had an incident at time t_r . The problem of interest in this study is the prediction of the congestion propagation patterns through the K -hop incoming neighbors of e_0 . Here, to predict congestion propagation patterns, we determine the propagation paths for all paths coming into e_0 , as defined in Definition 5.

Formally, given a time interval T during which congestion propagates and $\forall P_w = (e_0 e_{p_1}^w e_{p_2}^w \dots e_{p_K}^w) \in P^K(e_0)$, we predict $p\text{-seq}(P_w, T)$ as

$$\widehat{p\text{-seq}}(P_w, T) = (\hat{q}(e_{p_1}^w, T), \hat{q}(e_{p_2}^w, T), \dots, \hat{q}(e_{p_K}^w, T)), \quad (1)$$

where $\hat{q}(e_{p_i}^w, T)$ is the predicted value of $q(e_{p_i}^w, T)$. We predict $q(e_{p_i}^w, T)$ because the actual time interval T contains a future time span, which makes our problem difficult.

According to Definition 5, $q(e_{p_i}^w, T)$ indicates whether $e_{p_i}^w$ is propagated in T and can be obtained as $c(e_{p_i}^w, T)$. Consequently, to predict $q(e_{p_i}^w, T)$, we must predict $c(e_{p_i}^w, T)$. Therefore, we propose a prediction method for predicting $c(e_{p_i}^w, T)$ using historical speed data on e_0 . Additionally, we discuss how to determine an appropriate time interval T , which is referred to as the *propagation window* (PW) in this paper.

C. Similar pattern searching

To predict the $c(e_{p_i}^w, T)$ caused by a traffic incident on e_0 , we must search for traffic patterns that are similar to the non-recurrent traffic patterns generated by the incident. To search for traffic patterns, it is important to determine a reasonable time window for capturing traffic patterns. The time window is referred to as the *searching window* (SW) in this paper,

$$SW = [t_r - L : t_r]. \quad (2)$$

Let $s(e_0, [t_r - L : t_r])$ denote a *target traffic pattern*. Given the historical speed data on e_0 , $s(e_0, [1 : t_r])$, we wish to find similar patterns with the same time interval. Let $s(e_0, [t - L : t])$ and $t = L + 1, L + 2, \dots, t_r - L$ represent the candidates of similar patterns in the historical speed data, which are referred to as *source traffic patterns*. To define the similarity between a source traffic pattern and target traffic pattern, we adopt DTW [28], which has been widely used to measure the similarity between pairs of time series. Let $dtw(\cdot, \cdot)$ denote the DTW distance functions. Then, given a source traffic pattern and target traffic pattern, the DTW distance can be obtained as

$$dtw(s(e_0, [t_r - L : t_r]), s(e_0, [t - L : t])).$$

Therefore, we can find $s(e_0, [t^* - L : t^*])$, which is the most similar source traffic pattern to the target traffic pattern $s(e_0, [t_r - L : t_r])$, by solving the following optimization problem:

$$t^* = \arg \min_{t=L+1}^{t_r-L} dtw(s(e_0, [t_r - L : t_r]), s(e_0, [t - L : t])). \quad (3)$$

However, congestion propagation prediction based on the most similar source traffic pattern is inevitably limited in terms of accuracy and robustness. Therefore, we consider selecting the N most similar source traffic patterns. However, selecting the top- N source traffic patterns is difficult because many locally similar traffic patterns exist around the most similar patterns. For example, $s(e_0, [(t^* - L + 1) : (t^* + 1)])$ may have a near-optimal similarity distance compared to $s(e_0, [t^* - L : t^*])$. Such locally similar traffic patterns contain no additional information for congestion propagation prediction.

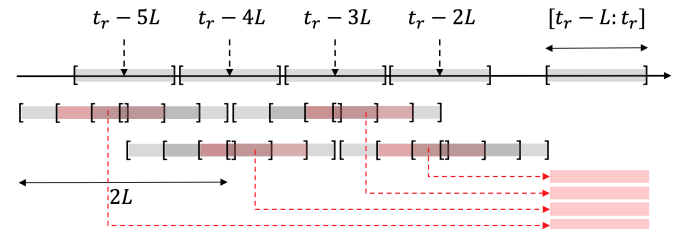


Fig. 2. Grouping search algorithm

To exclude locally similar patterns from consideration effectively, we introduce a grouping search algorithm. The concept of the grouping search algorithm is illustrated in Fig. 2. One

can see that the historical speed data on e_0 , $s(e_0, [1 : t_r])$ are split into multiple non-overlapping time periods

$$\begin{aligned} & \{[t_r - 2L : t_r - L], \dots, \\ & [t_r - (m+1)L : t_r - mL], \\ & \dots, [1 : 1 + L]\}. \end{aligned} \quad (4)$$

Suppose that there are M groups of non-overlapping time periods with lengths of $2L$ in Eq.(4). Then, for the m th non-overlapping time period, we apply Eq.(3) and obtain the most similar source traffic pattern within the time period as follows:

$$t^{*(m)} = \arg \min_{t=t_r-mL}^{t_r-(m+1)L} dtw(s(e_0, [t_r - L : t_r]), s(e_0, [t - L : t])). \quad (5)$$

Algorithm 1 Grouping search algorithm

Input: $t_r, L, N, s(e_0, [1, t_r])$

Output: $\mathcal{S}$

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1: for  $m = 1 : M$  do
2:    $\mathcal{D} \leftarrow \emptyset, group.dist, t^*$  // Initialization
3:   for  $t = (t_r - (m+1)L) : (t_r - mL)$  do
4:     if  $c(e_0, [t - L : t]) == 1$  then
5:        $dist \leftarrow dtw(s(e_0, [t_r - L : t_r]), s(e_0, [t - L : t]))$ 
// calculate DTW distances
6:       if  $dist < group.dist$  then
7:          $group.dist \leftarrow dist$ 
8:          $t^* \leftarrow t$ 
9:       end if
10:    end if
11:  end for
12:   $\mathcal{D} \leftarrow \mathcal{D} \cup \{t^* : group.dist\}$ 
// dictionary item={key:value}
13: end for
14:  $\mathcal{D}' \leftarrow sort(\mathcal{D})$  // sort  $\mathcal{D}$  in ascending order by value
15: return  $\mathcal{D}'.key[1 : N]$  // return keys of the first  $N$  items

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Based on Eq.(5), the grouping search algorithm yields a set of N time points

$$\{t^{*(n)}\}_{n=1}^N, \quad (6)$$

where $t^{*(n)}$ is the time point at which $s(e_0, [t^{*(n)} - L : t^{*(n)}])$ is the source traffic pattern that is the n th most similar to the target traffic pattern among $\{t^{*(m)}\}_{m=1}^M$. The grouping search algorithm is summarized in Algorithm 1.

There are three major traffic phases in congestion traffic flow: ‘free-flow phase’, ‘synchronized phase’, ‘congested phase’ [29]. The synchronized phase is a transitional traffic phase in which the flow of vehicles increases and the speed of them decreases before a congestion occurs. Particularly, for a non-recurrent congestion caused by a traffic incident, the synchronized phase indicates the occurrence of the traffic incident [30]. These phases reflect how the traffic pattern varies from the normal state to the state of being affected by the incident. Therefore, it can be stated that the ideal time window $[t_r - L : t_r]$ covers all three phases. Alternatively, L should be set to ensure coverage.

Because the appropriate length of L can be different for each traffic incident, we redefine Eq.(2) using the incident occurrence time. Let t_a denote the time point at which an incident occurs. Then,

$$SW = [t_r - L : t_r] = [t_a - PT : t_r], \quad (7)$$

where PT is a user-specific hyper-parameter that represents the length of the free-flow phase. A method of predicting t_a is proposed in the following section.

D. Propagation window

To predict $c(e_{pi}^w, T)$, we must rigorously determine the appropriate time window T . To this end, we define the propagation window PW as the time span over which the congestion caused by an incident e_0 is expected to propagate to the incoming roads $e_{pi}^w \in \{N_{in}^k(e_0)\}_{k=1}^K$. Then,

$$PW = [t_a : t_r + F]$$

is the time span from t_a to $t_r + F$, where F is a user-defined hyper-parameter, future time range to know the pattern of congestion propagation after reporting an incident.

However, PW is difficult to compute because t_a is unknown or cannot be specified accurately at time t_r . It takes approximately 15 min on average for information to be released after a traffic incident occurs in South Korea. However, this period can vary significantly depending on various circumstances.

Therefore, we propose a novel data-driven method to detect incident occurrence times. Suppose that a traffic incident occurs on e_0 and is reported at time t_r . Then, we detect the incident occurrence time t_a as

$$\hat{t}_a = \min\{t \in [t_r - 120 : t_r - 1] : t \text{ satisfies } (C1), (C2), (C3)\}, \quad (8)$$

where

$$(C1) : \bar{s}(e_0, t)_5 \leq \text{avg}(s(e_0, [t_r - 120 : t_r - 1])) \quad (9)$$

$$(C2) : \bar{s}(e_0, t)_5 \leq \bar{s}(e_0, t)_{30} \quad (10)$$

$$(C3) : s(e_0, t) \leq s(e_0, t - 1) - \text{sd}(s(e_0, [t - 10 : t])). \quad (11)$$

The first condition in Eq.(9) indicates that $\bar{s}(e_0, t)_5$ should be smaller than the average speed for the given time duration $[t_r - 120 : t_r - 1]$. The second condition in Eq.(10) is closely related to the concept of a death cross in stock chart analysis [31], [32] and is used to capture short-term speed patterns. It is known that a death cross appears on a chart when the moving average of short-term stock price crosses the long-term moving average, indicating that the trend of recent price drops is stronger than that of long-term price drops. We set the second condition to determine whether the short-term (5 min interval) speed trend overtakes the long-term (30 min interval) speed trend. The final condition in Eq.(11) sets a condition based on whether there is a radical speed drop. If a traffic incident occurs, the speed of the vehicles affected by the incident is expected to drop sharply after the incident occurs. To reflect variations in the speed distribution, if the speed decreases by

more than its standard deviation $\text{sd}(s(e_0, [t-10 : t]))$, then we detect a sharp speed drop.

By using the detected incident occurrence time $\hat{t}_a$ in Eq.(8), PW can be rewritten as

$$PW = [\hat{t}_a : t_r + F] = [t_r - (t_r - \hat{t}_a) : t_r + F]. \quad (12)$$

The detection algorithm for the incident occurrence time is summarized in Algorithm 2.

Algorithm 2 Incident occurrence time detection

Input: $t_r, s(e_0, [t_r - 120 : t_r - 1])$

Output: t_a

```

1:  $t_a \leftarrow t_r$  // Initialization
2: for  $t = (t_r - 120) : (t_r - 1)$  do
3:   if  $t$  satisfies  $(C1), (C2), (C3)$  then
4:     if  $t < t_a$  then
5:        $t_a \leftarrow t$ 
6:     end if
7:   end if
8: end for
9: return  $t_a$ 

```

E. Propagation prediction

For the N most similar source traffic patterns, we define $PW^{(n)}$ using $\{t^{*(n)}\}_{n=1}^N$ as follows:

$$PW^{(n)} = [t^{*(n)} - (t_r - \hat{t}_a) : t^{*(n)} + F],$$

where $\hat{t}_a$ is the predicted incident occurrence time from Eq.(8). It should be noted that $PW^{(n)}$ has the same length as PW .

Recall our problem statement in Section III-B. It is of interest to predict $c(e_{p_i}^w, PW)$, but this is difficult because the speed data for PW are not available at time t_r . Therefore, we propose a method for predicting $c(e_{p_i}^w, PW)$ using historical speed data for $PW^{(n)}$, $n = 1, \dots, N$.

For each $e_{p_i}^w \in \{N_{in}^k(e_0)\}_{k=1}^K$, we can obtain $c(e_{p_i}^w, PW^{(n)})$, $n = 1, \dots, N$,

$$c(e_{p_i}^w, PW^{(n)}) = \begin{cases} 1, & \text{if } e_{p_i}^w \text{ is congested in } PW^{(n)} \\ 0, & \text{otherwise,} \end{cases}$$

according to Definition 4. $e_{p_i}^w$ is congested in $PW^{(n)}$ if $\exists t \in PW^{(n)}$ such that

$$\max s(e_{p_i}^w, [t-10 : t]) < 0.7 \times ff(e_{p_i}^w),$$

where $ff(e_{p_i}^w)$ is the maximum speed restriction of $e_{p_i}^w$.

Once $c(e_{p_i}^w, PW^{(n)})$, $n = 1, \dots, N$, are obtained, we can predict $c(e_{p_i}^w, PW)$ as

$$\hat{c}(e_{p_i}^w, PW) = \begin{cases} 1, & \text{if } \frac{1}{N} \sum_{n=1}^N c(e_{p_i}^w, PW^{(n)}) > 0.5 \\ 0, & \text{otherwise.} \end{cases} \quad (13)$$

The procedure in Eq.(13) is repeated for $w = 1, \dots, W$, $i = 1, \dots, K$

For a path $P_w = (e_0 e_{p_1}^w e_{p_2}^w \dots e_{p_K}^w) \in P^K(e_0)$, $\hat{c}(e_{p_i}^w, PW)$, $i = 1, \dots, K$, yield the predicted sequence of propagation indicators, as defined in Eq.(1).

$$\widehat{p\text{-seq}}(P_w, PW).$$

After obtaining the predicted sequences of propagation indicators for all paths in $P^K(e_0)$, we define a *propagation matrix* as

$$\widehat{PM} := \begin{bmatrix} \widehat{p\text{-seq}}(P_1, PW) \\ \widehat{p\text{-seq}}(P_2, PW) \\ \vdots \\ \widehat{p\text{-seq}}(P_W, PW) \end{bmatrix}. \quad (14)$$

IV. EXPERIMENTAL RESULTS

The performance of the proposed method is evaluated using 215 real traffic incidents and related user trajectory data obtained from one of the major GPS navigation systems in South Korea. Experiments code is developed by Python 3.8 on AMD Ryzen Threadripper 3970X 32-Core processor workstation with Nvidia Titan RTX 24GB.

A. Dataset



Fig. 3. Traffic incidents location in Seoul (from September 2 to December 1, 2020)

Our traffic datasets consist of 1) a traffic road network dataset, 2) a traffic incident dataset within the traffic road network, and 3) a trajectory dataset of users using the GPS navigation system over a specific period of time.

The traffic road network dataset is for a major metropolitan area in the city of Seoul, South Korea, which contains nearly half of the entire country's population.

The incident dataset used in this study is provided by the Korea National Police Agency and contained 215 incidents that occurred from September 2 to December 1 of 2020 in Seoul, Korea. The red circles in Fig. 3 represent the locations of the 215 traffic incidents considered in this study. Let $e_{0,u} \in E$, $u = 1, \dots, 215$, denote the incident roads, described by both geographical coordinates and a road segment ID. Fig. 4(a) illustrates the histogram of $P^3(e_{0,u})$, which is the number of paths within 3-hop incoming neighbors. Each $e_{0,u}$ is associated with a reported time $t_{r,u}$. Fig. 4(b) and Fig. 4(c)

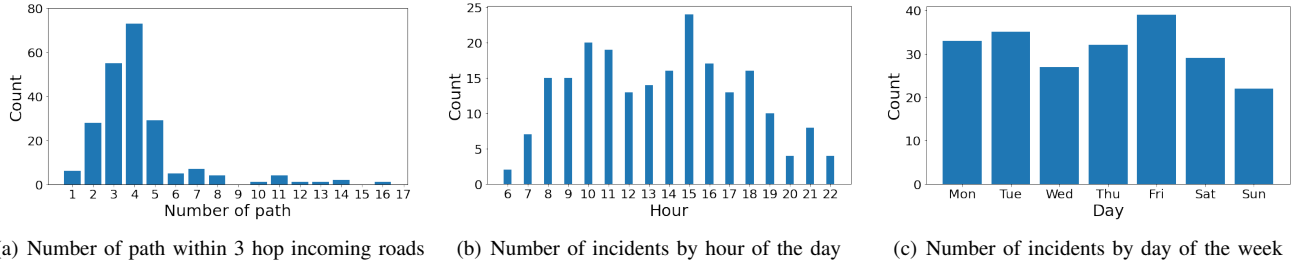


Fig. 4. Summary of the traffic incident dataset

show the distribution of hour of the day and day of the week at the time of occurrence, $t_{r,u}$, respectively.

The trajectory dataset consists of a series of points with latitude, longitude, and timestamp information generated when vehicles travel with navigation systems. The user trajectory data is converted into average speed data, $\{s(e, t) \in S : E \times T \rightarrow \mathbb{R}^+\}$, for all road segment in Seoul, $e \in E$, over the given time, $t \in T$, through a map matching process [33].

B. Experimental setup

Over the given time period, the speed data for all roads are normalized using their maximum speed restriction. For each traffic incident $e_{0,u}$, we applied the proposed method and predicted sequences of propagation indicators for all paths in $P^3(e_{0,u})$, $u = 1, \dots, 215$, and obtained the corresponding propagation matrix in Eq.(14).

The performance of the propagation prediction is evaluated by accuracy, recall, and precision. Under our experimental setup, accuracy is defined as

$$Accuracy = \frac{1}{215} \sum_{u=1}^{215} ACC_u, \quad (15)$$

where

$$ACC_u = \frac{\sum_{i=1}^3 \sum_{e_j \in N_{in}^i(e_{0,u})} \mathbb{1}(q(e_j, PW) = \hat{q}(e_j, PW))}{\sum_{i=1}^3 |N_{in}^i(e_{0,u})|}.$$

Recall and precision are defined in a similar way. Using these classification evaluation metrics, Section IV-D carried out a comparison study with the existing benchmarks and Section IV-E applied the proposed propagation prediction in real-time scenarios.

C. Hyper-parameter optimization

The proposed method includes user-specific hyper-parameters such as N in Eq.(6), PT in Eq.(7), and F in Eq.(12). For hyper-parameter optimization, we performed a grid search in the following hyper-parameter search space:

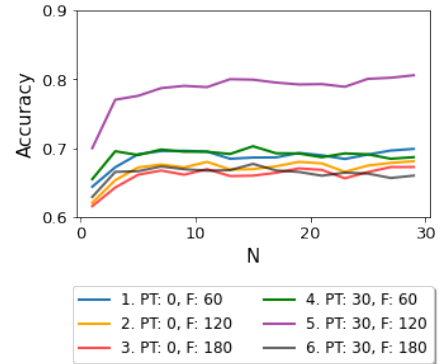
$$N = \{1, \dots, 30\}, PT = \{0, 30\}, F = \{60, 120, 180\}.$$

Table I and Fig. 5(a) show the results of hyper-parameter optimization based on accuracy in Eq.(15). As a result, the combination of $PT = 30$ and $F = 120$ outperforms the others, which is the purple solid line in the figures. For the combination of $PT = 30$ and $F = 120$, we observed that

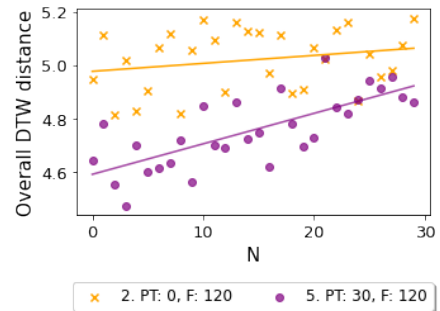
TABLE I
RESULTS OF A GRID SEARCH IN THE HYPER-PARAMETER SEARCH SPACE

	$PT = 0$			$PT = 30$		
	$F = 60$	120	180	$F = 60$	120	180
$N \leq 10$	0.696	0.677	0.668	0.698	0.790	0.674
$N \leq 20$	0.696	0.680	0.671	0.703	0.800	0.677
$N \leq 30$	0.699	0.681	0.673	0.703	0.806	0.677

the value of accuracy increases rapidly until $N = 5$ and then increases gradually after that. Hence, we can stress that 10 ~ 20 similar historical source traffic patterns are sufficient for predicting future congestion propagation patterns while maintaining computational efficiency and robustness.



(a) Relationship between N and Accuracy



(b) Overall DTW distance comparison for incident road

Fig. 5. Results of hyper-parameter optimization

Recall that PT represents the length of the free-flow phase in the searching window (SW). By adding traffic patterns in

the free-flow phase to the SW , we expect to be able to detect more similar source traffic patterns than when they are not added. To confirm the effect of adding PT , we compared the average DTW distances of 215 incidents between $PT = 0$ and $PT = 30$. Fig. 5(b) shows the average DTW distances for $N = 1, \dots, 30$ with $F = 120$. Consequently, the average DTW distances with $PT = 30$ outperformed the one with $PT = 0$, which means that the SW with the free-flow phase is effective to detect similar source traffic patterns.

D. Model comparison

The performance of the proposed method is compared with comparable benchmarking methods using 215 real traffic incidents and related user trajectory data. The hyper-parameters are set at $PT = 30$, $F = 120$, and $N = 13$.

We considered three existing benchmarks: traffic flow pattern classification using K-means clustering [34], short-term traffic forecasting using multivariate autoregressive models [35], and speed prediction using Graph WaveNet [36].

Basically, all benchmarks were trained in the experimental setup as explained in Section IV-B. For the traffic flow pattern classification method, we used mini-batch K-means with $K = 10$ by using scikit-learn¹. For the short-term traffic forecasting method, we implemented VAR algorithm by using statmodel². For the speed prediction method, the graph convolution network (GCN) was implemented with epoch of 10, batch size of 64, and a learning rate of 0.001 by using Pytorch package³.

TABLE II
PERFORMANCE COMPARISON

Models	Accuracy	Recall	Precision	Time(s)/ Incident
DTW	0.800	0.813	0.895	23.955
K-means	0.563	0.501	0.891	24.247
VAR	0.755	0.688	0.951	13.270
Graph Wavenet	0.714	0.611	0.768	102.474

As shown in Table II, the proposed pattern matching method outperformed all three benchmarks in terms of accuracy and recall values. The traffic forecasting method and the speed prediction method performed better than the pattern classification method in general, but they were poor in recall. It should be noted that recall values are more important in the context of congestion propagation prediction. The poor recall values of the existing methods might be due to inefficient information retrieval from non-recurrent traffic incidents.

E. Real-time experiments

We applied the proposed method in real-time scenarios. Suppose that the latest speed data comes after t_r and we want to update the proposed model with the new data. Even for real-time scenarios, the proposed method is suitable because

only the searching window in Eq.(7) needs to be modified as follows.

$$SW = [\hat{t}_a - PT : t_r + RU]. \quad (16)$$

TABLE III
PREDICTION RESULTS IN REAL-TIME SCENARIOS WITH $PT = 30$,
 $F = 120$, $N = 13$

Metric	$RU = 0$	$RU = 10$	$RU = 20$
Accuracy	0.800	0.803	0.816
Recall	0.813	0.826	0.840
Precision	0.895	0.889	0.882

Table III showed the performance of the proposed model with various values of RU in Eq.(16). As expected, the accuracy values increased in the following order: 0.792, 0.801, and 0.811 as the value of RU increased. Similarly, the recall values increased in the order of 0.802, 0.828, and 0.836. This was because the broader SW reflected more information about non-recurrent traffic patterns on the incident roads. In the process of matching the target traffic pattern with the historical source traffic patterns, the more similar source traffic patterns were selected, which resulted in the more accurate propagation prediction.

V. CONCLUSION

In this paper, we presented a novel online learning method based on pattern matching to predict the propagation patterns of non-recurrent traffic congestion. Also, we developed a grouping search algorithm for effective pattern matching. By using pattern matching techniques, our model effectively matched similar historical speed pattern of an incident road and predicted the future non-recurrent congestion propagation patterns with high accuracy and recall values. This results demonstrated the hypothesis that the current non-recurrent propagation pattern evolves similarly to the historical recurrent pattern. Furthermore, we validated the applicability to real-time scenarios by improving the performance of the model as the relevant information is updated in real-time.

Unlike traditional machine learning methods or deep learning methods that are focused on recurrent patterns and require pre-defined traffic network and corresponding parameters, the proposed method can be applied to any incident anytime, anywhere. This opens up a new direction in modelling non-recurrent congestion propagation patterns.

In future work, we will try to match patterns using other traffic indicators such as volume, flow, and density that can explain the accident situation and explore approaches to learn dynamic nature of non-recurrent congestion propagation patterns.

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¹<https://github.com/scikit-learn/scikit-learn>

²<https://github.com/statsmodels/statsmodels>

³<https://github.com/nznzhan/Graph-WaveNet>

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